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Submitter Information

Name: H. Chad Lane

General Comment

Please see attached document.

Attachments

Public Comment on National AI Strategy and Priorities

Public Comment on National AI Strategy and Priorities

Submitted by:

- H. Chad Lane, Professor of Educational Psychology & Computer Science, University of Illinois, Urbana-Champaign
- Jamie Payton, Dean, Ying Wu College of Computing, New Jersey Institute of Technology

Introduction

As artificial intelligence continues to transform our society, economy, and national security, the United States must establish a comprehensive and forward-looking strategy to maintain global leadership in AI research, development, and implementation. We respectfully submit the following recommendations to inform the Administration's AI policy priorities, with particular emphasis on education, research funding, and workforce development.

Policy Recommendations

1. Expand NSF's AI in K-12 Education Programs

The National Science Foundation has demonstrated remarkable foresight by investing in AI K-12 education for well over a decade. In light of recent advances in generative AI and increased international competition, we recommend significantly expanding these successful NSF initiatives. This expansion should build upon existing frameworks while updating curricula to address emerging AI capabilities and applications. The program should continue to support the development of age-appropriate materials that address four essential levels of AI engagement: conceptual understanding, practical awareness, application skills, and systems control. Expanded funding should support teacher professional development, curriculum materials, and necessary technology infrastructure to ensure all American students develop AI literacy from an early age, matching or exceeding initiatives in countries like China that have alreadyimplemented systematic AI education beginning in elementary school.

2. Strengthen NSF Funding for AI Research and Education

The National Science Foundation has been instrumental in maintaining U.S. leadership in scientific and technological innovation. We recommend significantly increasing NSF funding specifically designated for AI research, with particular emphasis on:

- Foundational AI research that advances capabilities in machine learning, natural language processing, and multimodal systems
- Use-inspired research that applies AI to critical domains including education, healthcare, and scientific discovery
- Research on effective human-AI collaboration models to maximize productivity and innovation
- Development of secure, privacy-preserving AI technologies that protect user data

This funding should prioritize both individual research grants and the establishment of multi-institutional research centers that bring together diverse expertise to address complex challenges. Growth in these areas has had and could continue to have a profound impact on American innovation, commercial growth, and to maintain global leadership in AI. Our advantages on all fronts are significant.

3. Create an AI-Ready Workforce Development Program

To prepare Americans for the jobs of tomorrow, we recommend establishing a coordinated workforce development program that spans secondary education, higher education, and mid-career training. This program should:

- Support the development of undergraduate and graduate programs in AI and related fields
- Fund apprenticeship and certification programs that provide practical AI skills training
- Create incentives for industry-academic partnerships that connect education to real-world applications
- Establish portable credentials that validate AI competencies acquired through various educational pathways

4. Support Research on AI-Human Collaboration Models

The most promising future for AI lies not in replacing human capabilities but in creating powerful collaborative relationships between humans and AI systems. We recommend dedicated funding for research that explores optimal models of AI-human collaboration across critical sectors. This research should investigate how AI can:

- Augment teacher capabilities through enhanced awareness of student needs
- Support personalized learning approaches that adapt to individual student progress
- Enhance workforce productivity in blue-collar professions like automotive repair, construction, and manufacturing
- Improve healthcare delivery through AI-assisted diagnostics and treatment planning
- Strengthen customer service while maintaining meaningful human interaction

- Create new employment opportunities rather than displacing workers

5. Develop AI Integration for National Security and Defense

AI is radically changing how we can protect ourselves and our allies, and our national security increasingly depends on both our service members and civilians understanding AI technologies. To maintain technological superiority in defense capabilities, we recommend establishing a strategic initiative to responsibly integrate AI systems across military training, operations, and logistics. This initiative should:

- Fund research on military-specific AI applications that enhance operational effectiveness, both cyber and physical
- Develop AI-based training systems that improve military readiness through realistic simulations
- Create secure, robust AI systems that can operate in contested environments
- Establish guidelines for responsible AI use in national security contexts
- Ensure broad AI literacy among both service members and the civilian workforce supporting defense

6. Strengthen U.S. Higher Education Leadership in AI

America's higher education system is a powerful engine for innovation and talent development. To maintain this competitive advantage, we recommend policies that:

- Increase graduate fellowship funding in AI-related fields to attract and retain top talent
- Support AI research infrastructure at universities including computational resources and datasets
- Facilitate knowledge transfer between academic research and commercial applications
- Expand international student programs in AI fields while creating pathways to retain this talent in the U.S.
- Establish AI-themed pipelines between K-12 education and higher education to ensure that talent is grown within the U.S. and remains here

7. Create a National AI Testing and Evaluation Framework

To ensure AI systems meet standards for reliability, safety, and performance, we recommend developing a comprehensive testing and evaluation framework that:

- Establishes metrics and benchmarks for AI system performance across various domains
- Creates standardized evaluation protocols for key AI capabilities
- Provides resources for independent verification and validation of AI systems
- Supports research on methods to test increasingly complex AI architectures

8. Fund Foundational AI Research for National Benefit

The federal government should prioritize funding for AI research that serves as a foundation for national advancement but may not occur naturally through industry or state-level initiatives. Such funding should focus on:

- Basic research in AI that may not have immediate commercial applications but will drive long-term innovation and discovery
- Educational applications of AI that support learning and development across the lifespan
- Capacity-building infrastructure that democratizes access to AI tools and knowledge
- Empirical research that objectively evaluates AI impacts, benefits, and potential risks
- Cross-disciplinary initiatives that connect AI capabilities with humanities, social sciences, and other fields

This research is most effectively conducted through our world-leading universities and research institutions, which have consistently demonstrated their ability to advance fundamental knowledge while training the next generation of AI innovators and leaders. By strategically investing in this foundational research, the United States will maintain its global leadership position while ensuring AI development aligns with broader societal goals and values.

Conclusion

The United States stands at a critical juncture in AI development and implementation. With strategic federal investment and coordinated policy initiatives, America can maintain its global leadership in AI innovation while ensuring these technologies benefit all citizens. The recommendations outlined above represent essential steps toward a comprehensive national AI strategy that balances technological advancement with thoughtful implementation and broad access to AI literacy and skills.

By investing in research, education, and workforce development, the United States can ensure that AI technologies enhance human capabilities, strengthen our economy, and support our national security while reflecting American values and priorities. Public understanding of AI, widespread access to resources, and provision of cutting-edge AI tools to American families are critical components of maintaining America's technological leadership. Our higher education system and research universities, supported by robust federal funding, will continue to serve as the cornerstone of American leadership in AI innovation and application, ensuring these benefits reach all sectors of society.